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SUPERSEDING
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MILITARY SPECIFICATION

BUCKLE, CENTER BAR (MILITARY POLICE BELT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers center bar buckles.

1.2 Classification. The buckles shall be of the following classes as specified (see 6.2).

Class 1 - Solid polished brass (Marine Corps)
Class 2 - Die cast zinc and aluminum alloy with chrome plated finish
Class 3 - Die cast zinc and aluminum alloy with brass plated finish

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8315

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SPECIFICATIONS

FEDERAL

QQ-B-613	-	Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-W-321	-	Wire, Copper Alloy
QQ-Z-363	-	Zinc-Base Alloy; Die Castings
PPP-B-26	-	Bag, Plastic, (General Purpose)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-665	-	Boxes, Paperboard, Metal Edged and Components
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

4-1-73	-	Buckle, Center Bar (Military Police Belt)

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the test of this document and the references cited herein, the test of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.2.1 Brass, stamping. The brass for stamping the class 1 buckle body shall conform to alloy number 260 or 268, quarter-hard temper of QQ-B-613.

3.2.2 Wire, copper alloy. The wire for the tongue for all classes of buckles shall conform to alloy number 260 and 270, quarter-hard temper of QQ-W-321.

3.3 Design and construction. The design construction of the buckle shall conform to Drawing 4-1-73 and the requirements specified herein.

3.3.1 Body.

3.3.1.1 Class 1 buckle. The body for the class 1 buckle shall be stamped from the brass specified in 3.2.1. A cutout shall be provided in the center bar to retain the tongue in position in the center of the bar and also so that the total lateral movement of the tongue tip meets the requirement specified in 3.3.2. The body shall be free from cracks, burrs, drag, step and rough or sharp edges.

3.3.1.2 Class 2 and 3 buckle. The body for the class 2 and class 3 buckles shall be die cast from a zinc base alloy conforming to alloy AG40A of QQ-Z-363. Ribs shall be provided on the center bar to retain the tongue in

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the center of the bar and also so that the total lateral movement of the tongue tip meets the requirement specified in 3.3.2 The body shall be free of rough or sharp edges, pits, pores, blowholes and cracks.

3.3.2 Tongue. The tongue shall be fabricated from the wire specified in 3.2.2. After assembly to the buckle body the tongue shall not bind on the center bar, the tip of the tongue shall not protrude beyond the outer edge of the body but shall protrude a minimum of 3/64 inch beyond the inner edge of the body, it shall not be possible by hand to slide the tongue out of the cutout in the center bar of the class 1 body or over the ribs in the center bar of the class 2 and class 3 body, and the total lateral movement of the tongue tip shall not exceed 5/8 inch.

3.4 Finish. The buckle body and tongue shall be finished as specified below (see 4.4.1.1). Prior to finishing, the buckle body and tongue shall be thoroughly cleaned in accordance with commercial practice and shall be free from corrosion, dirt or discoloration.

3.4.1 Tongue, class 1 and class 3 buckle. The tongue for the class 1 and class 3 buckles shall be brass dipped. The tongue for the class 3 buckle shall be coated with a clear baked on epoxy lacquer. When dry, the lacquer shall adhere tightly, shall not be tacky, and shall be free of wrinkles, sags, runs, pinholes, blushing, orangepeel, dirt, dust or other foreign matter.

3.4.2 Tongue, class 2 buckle. The tongue for the class 2 buckle shall be nickel plated 0.0001 $\pm$ 0.000025 inch thick followed by coating with lacquer the same as specified in 3.4.1.

3.4.3 Body, class 1 buckle (stamped). The top surfaces of the class 1 buckle body exposed to view when worn shall be polished to a uniform mirror like finish.

3.4.4 Body, class 2 buckle (die cast). All surfaces of the class 2 buckle body shall be burnished to a smooth surface texture. The buckle body shall then be copper plated 0.0003 $\pm$ 0.00005 inch thick followed by nickel plating 0.0002 $\pm$ 0.00005 inch thick followed by chrome plating not less than 0.000015 inch thick. The platings shall be free of frosty areas, blisters, pimples or peeled or cut through areas. The top surfaces of the buckle body exposed to view when worn shall be polished to a uniform mirror like finish.

3.4.5 Body, class 3 buckle (die cast). The class 3 buckle body shall be prepared and plated the same as specified for the class 2 buckle body except in lieu of chrome plating the final plating shall be brass not more than 0.00005 inch thick. After polishing, the buckle body shall be coated with a clear baked on epoxy lacquer. When dry, the lacquer shall adhere tightly, shall not be tacky, and shall be free of wrinkles, sags, runs, pinholes, blusing, orange-peel, dirt, dust and other foreign matter. The lacquer coating shall not flake when tested as specified in 4.5.

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3.5 Marking for identification. All buckles shall be identified with the manufacturer's name or trademark. The marking shall be permanently affixed by stamping, etching or casting on the back of the buckle and shall be legible. The marking shall be incorporated in such a manner that the surface opposite the marking shows no sign of penetration or unevenness, and the distortion is not caused on any part of the buckle.

3.6 Workmanship. The finished buckle shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article. When a first article is required (see 3.1), it shall be examined as specified in 4.4.3.1 and 4.4.3.2. The presence of any defects shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified and qualified in this document or applicable purchase document.

4.4.1.1 Certificate of compliance. The contractor shall furnish a certificate of compliance certifying that the plating and lacquer coating on the buckle body and tongue where applicable conform to the specified requirements in 3.4.

4.4.2 In-process inspection. In-process examination shall be made to assure that prior to finishing, the buckles are free of the defects described in 3.3.1.2 and 3.4. Whenever nonconformance is noted, correction shall be

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made to the applicable process and all items processed. Buckles which can not be corrected shall be removed from production.

4.4.3 End item inspection.

4.4.3.1 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of buckles. The sample unit shall be one buckle. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish (general)	Not polished to a mirror-like finish in areas as specified	X	
	Surface not smooth:		
	- on surfaces exposed to view in use position	X	
	- on surfaces not exposed to view in use position		X
	Discoloration, corrosion or stain clearly noticeable:		
	- on surfaces exposed to view in use position	X	
	- on surfaces not exposed to view in use position		X
	Abraded, scuffed, or scratched:		
	- on surfaces exposed to view in use position	X	
	- on surfaces not exposed to view in use position		X
	Not finished or the type of finish is not as specified	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Class 3 buckle assembly and tongue on class 2 buckle	Not lacquer coated or area of not not coating	X	
	Lacquer coating blushed, cloudy, or not clear		X
	Lacquer coating not adherent, i.e., blistered, flaking or peeling		X
	Foreign matter imbedded in lacquer coating clearly noticeable i.e., lint, dust or dirt:		
	- on surfaces exposed to view in use position	X	
	- on surfaces not exposed to view in use position		X
	Lacquer coating not continuous or not level, i.e., has run, sag, drop, pinholes, blushing or orange peel, or is wrinkled or tacky		X
Class 2 buckle body	Not chrome plated	X	
	Chrome plate has frosty area on surfaces exposed to view in use position	X	
	Plating not continuous, not adherent, or is blistered, pimpled, peeled or cut through	X	
	Foreign matter included	X	
Quality of metal (class 2 and class 3 buckles)	Surface spotted or open grained, i.e., pitted, porous or contains blowhole or crack	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Contruction and work-manship	Any mutilation such as dig, gouge, nick, dent, scratch or metal mark, clearly noticeable:		
	- on surfaces exposed to view in use position	X	
	- on surfaces not exposed to view in position		X
	Distorted, bent, malformed, cracked, drag or step	X	
	Burr, fin, rough or sharp edge	X	
	Tongue missing, broken, fractured, malformed, bent or damaged	X	
Assembly	Not properly assembled, i.e., tongue in reverse position	X	
	Tongue loop loose or open to an extent such that the tongue can be dislodged manually from its position on the center bar	X	
	Tongue binds on center bar		X
Marking	Not as specified, not legible or omitted		X

4.4.3.2 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on Drawing 4-1-73. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of buckles. The sample unit shall be expressed in units of buckles. The sample unit shall be one buckle of one class. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units. Shall be 4.0.

4.4.4 End item testing. The end item shall be tested as specified in 4.5.1. The sample unit shall be one buckle. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.4.5 Packaging inspection. Examination shall be made to determine that preservation, packing and marking comply with section 5. Defects shall be scored as set forth in table II. The sample unit shall be one shipping container fully packaged. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. Packaging defects

Examine	Defect
Markings (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Material	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of tape; bulged or distorted container
Contents	Number of intermediate packs per container is more or less than required
	Number of buckles per intermediate pack more or less than required 1/

1/ For this defect, two intermediate packs in the sample shall be examined.

4.4.6 Palletization examination. An examination shall be made to determine that palletization complies with the section 5 requirements. Defects shall be scored as indicated in table III. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. Palletization defects.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.

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TABLE III. Palletization defects. (cont'd)

Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.5 Method of inspection.

4.5.1 Lacquer test. The lacquer finish on the body of the buckle shall be tested by scratching the surface with a knife blade for a distance of not less than 1-inch. Any nonconformance with the flaking requirements of 3.4.5 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Leval A.

5.1.1.1 Unit pack. Each buckle shall be enclosed in a close fitting clear polyethylene bag conforming to type II, style 1 of PPP-B-26, except that the film thickness shall be 0.00125 (± 25 percent) inches.

5.1.1.2 Intermediate pack. Fifty buckles of like description, unit packed as specified in 5.1.1.1, shall be uniformly and compactly packed in a close-fitting intermediate paperboard box conforming to variety 1, style III, type G, class i, or style IV, type I or J, class i of PPP-B-566; style A of PPP-B-665; or type I, variety 1, class A, style 2 of PPP-B-676. Each box shall be closed with 2-inch minimum width tape conforming to type III, grade B of PPP-T-45, applied girthwise around the center and overlapping not less than 2 inches.

5.1.2 Commercial. The buckles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Five hundred buckles of like description, preserved as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

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5.2.2 Level B packing. Five hundred buckles of like description, preserved as specified in 5.1, shall be packed in a close fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Buckles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), cups, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM 3951, as applicable.

6. NOTES

6.1 Intended use. The center bar buckle is intended for use with the military police belt.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document
b. Class required (see 1.2)
c. When a first article sample is required (see 3.1, 4.3 and 6.3)
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2)
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1)
f. When palletization is required (see 5.3)

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample and should consist of one buckle assembly. The contracting officer should include specific instructions

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in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - MC

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL
Project No. 8315-0311

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